

Ordinal complexity of graph minors

Component splitting, degree-two graphs,
and discontinuous limits

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Abstract

The ordinal invariants of the graph-minor ordering provide a concrete entry point into an open program in well-partial-order theory. We study its restriction to finite simple graphs of maximum degree two, and to related classes with bounds on path size and the number of cyclic components. A component-splitting argument, combined with the established maximal-order-type theorem for finite multisets, proves that the finite disjoint-union closure of a connected graph class of maximal order type $\alpha < \varepsilon_0$ has maximal order type ω^α . In particular, the degree-two graph-minor order has maximal order type and ordinal width both equal to ω^{ω^2} , while its height is only ω . We give exact formulas throughout a two-parameter family, including $\omega^{\omega+k}$ when at most k cycles are allowed and path sizes are unbounded. These formulas yield explicit failures of continuity under increasing unions, even unions of minor-closed classes. An exact cycle-allocation and path-packing criterion is proved and checked against closure under elementary minor operations on all 1,500 isomorphism classes with at most fourteen vertices. The work is a partial attack on the general graph-minor problem, not a calculation for all finite graphs; historical novelty of the special-case formulas is not asserted.

Central calculation. For $\mathcal{D}$, the finite simple graphs of maximum degree at most two, ordered by minors,

$$\mathfrak{o}(\mathcal{D}) = \mathfrak{w}(\mathcal{D}) = \omega^{\omega^2}, \quad \mathfrak{h}(\mathcal{D}) = \omega.$$

The archive includes this manuscript, its L^AT_EX source, an independent finite checker, machine-readable results, and a proof audit.

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1 The selected problem and the scope of the attack

A well-partial-order is not determined, even approximately, by its cardinality. The coordinatewise orders on $\mathbb{N}$, $\mathbb{N}^2$, and $\mathbb{N}^3$ are all countable, have no infinite antichain, and have only finite descending sequences. Nevertheless, their maximal linear-extension types are ω , ω^2 , and ω^3 . The ordinal invariant records how much freedom remains when one resolves incomparabilities into a total order.

Vialard’s 2024 thesis identifies the ordinal measurement of finite undirected graphs under the minor relation as a promising unresolved direction [8, p. 110]. The broad question motivating this manuscript is the following.

Problem 1.1 (Graph-minor ordinal measurement). Determine the maximal order type, and associated ordinal invariants, of finite graphs under the minor relation, preferably in a form that interacts with labels and structural decompositions.

The present attack focuses on a class where the graph-theoretic geometry can be described exactly, while the resulting order types are already substantially transfinite. Every connected finite simple graph of maximum degree at most two is either a path or a cycle. Disconnected graphs in this class are therefore finite multisets of these components. The main obstacle is that *minor containment is not multiset embedding*: one source component can split into several target components.

For instance,

$$P_2 \dot{\cup} P_2 \preccurlyeq_m P_4, \quad P_2 \dot{\cup} P_3 \preccurlyeq_m C_5,$$

where P_n has n vertices and C_n is a simple n -cycle. Any argument that requires distinct source components for distinct target paths would miss these minor relations. Conversely,

$$C_3 \dot{\cup} P_1 \not\preccurlyeq_m C_4 :$$

retaining a cycle uses the entire cyclic source component, leaving no vertices for a separate path. Understanding both facts is essential.

1.1 Results obtained

For $b \in \{1, 2, 3, \dots, \infty\}$ and $k \in \{0, 1, 2, \dots, \infty\}$, let $\mathcal{D}_{b,k}$ consist of the graphs whose path components have at most b vertices and which have at most k cyclic components. The symbol ∞ means that the corresponding bound is absent; it is not an ordinal parameter. Cycle lengths are unrestricted throughout. Write $\mathcal{D} = \mathcal{D}_{\infty, \infty}$.

Theorem 1.2 (Complete two-parameter calculation). *With $b \geq 1$ and the preceding conventions, the invariants are as follows.*

Path-size bound	Cycle-count bound	$\mathfrak{o}(\mathcal{D}_{b,k})$	$\mathfrak{w}(\mathcal{D}_{b,k})$	$\mathfrak{h}$
$b < \omega$	$k < \omega$	ω^{b+k}	ω^{b+k-1}	ω
$b = \infty$	$k < \omega$	$\omega^{\omega+k}$	$\omega^{\omega+k}$	ω
$b < \omega$	$k = \infty$	ω^ω	ω^ω	ω
$b = \infty$	$k = \infty$	$\omega^{\omega \cdot 2}$	$\omega^{\omega \cdot 2}$	ω

The proof occupies sections 2 to 8. The result is not based on extrapolating finite computations.

A general component theorem is also proved: if C is a well-partial-order of connected finite simple graphs with $\mathbf{o}(C) = \alpha < \varepsilon_0$, then its finite disjoint-union closure has maximal order type ω^α . For a finite collection of $N \geq 1$ connected templates, the corresponding maximal order type and width are ω^N and ω^{N-1} , independently of the detailed minor relations between the templates.

1.2 What is and is not claimed

The standard facts about maximal linearizations, natural products, multisets, and the height–width inequality are attributed below. The component reduction and the degree-two calculations are given with proofs. They are applications and extensions of that established machinery, not a claim to have invented it. The literature search located the broad open program, but did not establish priority for these particular special cases. The formulas should therefore be read as explicit, auditable mathematical results derived in this investigation, not as a claim that a named longstanding conjecture has been resolved.

The manuscript does not determine $\mathbf{o}$ for all finite graphs, for all finite trees, or for edge-labeled graph-minor orders. It does identify where the present method is exact, gives a nontrivial parameter spectrum, and isolates a genuine obstruction to extending the same proof indiscriminately.

2 Ordinal conventions and the established tools

2.1 Three different ranks

We use $\mathbb{N} = \{0, 1, 2, \dots\}$. All graphs are considered up to isomorphism. For finite simple graphs, the minor relation on isomorphism classes is antisymmetric. A proper minor operation strictly decreases $|V| + |E|$, and mutual minor containment therefore forces isomorphism.

A sequence $(x_0, \dots, x_{r-1})$ in a partial order P is *bad* if $x_i \not\leq x_j$ whenever $i < j$. Let $\text{Bad}(P)$ be the tree of finite bad sequences, including the empty sequence. Define $\text{Ant}(P)$ similarly using pairwise incomparable entries, and $\text{Dec}(P)$ using strictly descending sequences. For a well-founded tree, the rank of a node is the supremum of the ranks of its immediate extensions plus one. Leaves have rank zero. A partial order is a *well-partial-order* when its bad-sequence tree is well-founded; equivalently, it has neither an infinite descending chain nor an infinite antichain.

Definition 2.1. The *maximal order type* $\mathbf{o}(P)$, *ordinal width* $\mathbf{w}(P)$, and *height* $\mathbf{h}(P)$ are the ranks of the empty sequence in $\text{Bad}(P)$, $\text{Ant}(P)$, and $\text{Dec}(P)$, respectively.

In particular, a nonempty chain has width 1, not 0, and a finite n -element antichain has width n . An ordinal width such as ω^{ω^2} is a rank of the tree of finite antichains; it is *not* the cardinality or order type of an actual antichain.

The residual identities are

$$\mathbf{o}(P) = \sup_{x \in P} (\mathbf{o}(\{y : x \not\leq y\}) + 1), \quad (1)$$

$$\mathbf{w}(P) = \sup_{x \in P} (\mathbf{w}(\{y : y \perp x\}) + 1), \quad (2)$$

$$\mathbf{h}(P) = \sup_{x \in P} (\mathbf{h}(\{y : y < x\}) + 1). \quad (3)$$

They follow directly by taking the first entry of a sequence.

2.2 Natural arithmetic

We write $\oplus$ and $\otimes$ for Hessenberg natural sum and product. Natural sum adds the coefficients in a common Cantor normal form. Natural product multiplies these normal forms as polynomials, using natural sum for exponents. Thus

$$(\omega^\xi) \otimes (\omega^\eta) = \omega^{\xi \oplus \eta}.$$

Ordinary ordinal addition and multiplication are written $+$ and $\cdot$. The distinctions needed below include

$$b + \omega = \omega, \quad b \oplus \omega = \omega + b, \quad \text{and} \quad \omega^\omega \cdot \omega^k = \omega^{\omega+k} \quad (b, k < \omega).$$

A finite natural sum of monomials with exponents strictly below ξ is below ω^ξ . Natural sum is strictly increasing in each argument. An infinite additively indecomposable ordinal is an ordinal ω^ξ with $\xi > 0$; adding finitely many elements to an ordinal strictly below it still leaves an ordinal strictly below it.

2.3 Explicit external dependencies

We use four established results. They are recorded here to make the proof dependencies transparent.

Theorem 2.2 (de Jongh–Parikh). *A well-partial-order has a linear extension whose order type is exactly $\mathfrak{o}(P)$. Moreover,*

$$\mathfrak{o}(P \times Q) = \mathfrak{o}(P) \otimes \mathfrak{o}(Q), \quad \mathfrak{o}(P \sqcup Q) = \mathfrak{o}(P) \oplus \mathfrak{o}(Q),$$

where $\sqcup$ denotes a disjoint union with no cross-comparabilities.

These are the linearization and composition results of [1]. In particular, $\mathfrak{o}(\mathbb{N}^d) = \omega^d$ for a positive finite d .

Theorem 2.3 (Finite multiset maximal order type). *Let P be a well-partial-order with $\mathfrak{o}(P) = \alpha < \varepsilon_0$. Order finite multisets by the existence of an injective matching that maps each entry to a greater or equal entry. Then*

$$\mathfrak{o}(\mathcal{M}_{\text{emb}}(P)) = \omega^\alpha.$$

Here ε_0 is the least nonzero ordinal satisfying $\omega^{\varepsilon_0} = \varepsilon_0$. We use only this below- ε_0 specialization of the multiset theorem [4, 5]; the general theorem has an exponent correction. A modern presentation distinguishes carefully between multiset embedding and multiset replacement [7].

Theorem 2.4 (Kříž–Thomas). *For every well-partial-order P ,*

$$\mathfrak{w}(P) \leq \mathfrak{o}(P) \leq \mathfrak{h}(P) \otimes \mathfrak{w}(P).$$

This is the height–width inequality of [3]. The last standard fact is closure of well-quasi-ordering under finite multisets; it follows from Higman’s theorem by forgetting the order of a finite word [2].

For clarity, the following elementary comparison rules are proved rather than left implicit. Passing to an induced suborder cannot increase any of the three tree ranks. Adding comparabilities on the same underlying set cannot increase $\mathfrak{o}$ or $\mathfrak{w}$: it only removes bad sequences and antichains. Finally, if P is partitioned into finitely many pieces P_i , then

$$\mathfrak{w}(P) \leq \bigoplus_i \mathfrak{w}(P_i), \quad \mathfrak{o}(P) \leq \bigoplus_i \mathfrak{o}(P_i). \quad (4)$$

For maximal order type, remove all cross-comparabilities and apply the disjoint-sum formula. For width, at an antichain-tree node take the natural sum of the antichain ranks remaining in the individual pieces. Extending the antichain lowers the rank in its chosen piece and does not increase any other summand. This is a strict ordinal ranking with the displayed root value, proving the upper bound. A finite cover can be made a partition by assigning each point to its first containing piece; each resulting piece is an induced suborder of the original covering piece.

3 A component-splitting transfer theorem

Let C be a collection of pairwise nonisomorphic, nonempty, connected finite simple graphs. Give it the inherited minor relation. Define

$$\mathcal{A}(C) = \{C_1 \dot{\cup} \cdots \dot{\cup} C_r : r < \omega, C_i \in C\},$$

including the empty graph when $r = 0$. Neither C nor $\mathcal{A}(C)$ is required to be minor-closed. The relation always means ordinary graph minor containment in the ambient class of all finite graphs; intermediate minor operations need not stay inside the displayed subclass.

Lemma 3.1 (Allocation to source components). *Suppose $H \preccurlyeq_m G$. Each connected component of H can be assigned to a unique connected component of G by a minor model. The target components assigned to a source component C are each connected minors of C . If one of them is isomorphic to C , it is the only target component assigned to that source component.*

Proof. A minor model assigns to each target vertex a nonempty connected branch set in the source, with distinct branch sets disjoint. Branch sets representing a connected target component, together with the edges realizing its adjacency, all lie in one source component. This gives the allocation and also a connected-minor model for each target component.

If a target component is isomorphic to C , it has $|V(C)|$ vertices. Its $|V(C)|$ nonempty disjoint branch sets already use every vertex of C . There is no vertex left to represent another target component. $\square$

Lemma 3.2 (Cantor-normal-form linearization). *Let $\ell : C \rightarrow \alpha$ be a bijection that realizes a well-order linear extension of the connected minor order. For a graph G in $\mathcal{A}(C)$, put*

$$\Phi_\ell(G) = \bigoplus_{C \text{ a component of } G} \omega^{\ell(C)}. \quad (5)$$

Then Φ_ℓ is a bijection from $\mathcal{A}(C)$ onto the ordinals below ω^α . Moreover,

$$H \preccurlyeq_m G, H \not\cong G \implies \Phi_\ell(H) < \Phi_\ell(G).$$

Consequently, comparing these codes gives a linear extension of type ω^α .

Proof. Unique component decomposition and uniqueness of Cantor normal form show that the map is injective. Conversely, every ordinal below ω^α is a finite Cantor-normal-form polynomial with exponents below α . For each exponent, take the graph component with that ℓ -value and use the required finite multiplicity. This proves surjectivity, including the empty graph and the ordinal zero.

Now allocate target components to source components as in lemma 3.1. If a source component C survives unchanged, its contribution is $\omega^{\ell(C)}$. Otherwise, it either produces no target component, or produces a finite list of proper connected minors D with $\ell(D) < \ell(C)$. Their total natural-sum contribution is strictly smaller than $\omega^{\ell(C)}$. In the case $\ell(C) = 0$, only the empty list can occur in this latter alternative. Thus each source contribution weakly decreases, and every changed source contribution strictly decreases. If none changes, the whole output is isomorphic to the input. Therefore a proper minor changes at least one contribution. Strict monotonicity of finite natural sums proves the asserted inequality. $\square$

Theorem 3.3 (Component transfer below ε_0). *If C is a well-partial-order under minors and $\mathbf{o}(C) = \alpha < \varepsilon_0$, then $\mathcal{A}(C)$ is a well-partial-order and*

$$\mathbf{o}(\mathcal{A}(C)) = \omega^\alpha.$$

Proof. Identify a graph with its finite multiset of connected components. An injective componentwise minor matching is sufficient for a graph minor model, since the corresponding models can be placed in distinct source components. Hence the minor order on $\mathcal{A}(C)$ is an augmentation of $\mathcal{M}_{\text{emb}}(C)$. This proves well-partial-ordering and, by theorem 2.3, gives the upper bound ω^α .

By theorem 2.2, the connected order has a linear extension of type α . Apply lemma 3.2 to obtain a linear extension of $\mathcal{A}(C)$ of type ω^α . This is the matching lower bound. $\square$

Remark 3.4 (Why splitting does not spoil the calculation). Componentwise matching is used only for the upper bound. The lower bound permits a source component to split, but charges every resulting proper connected minor below the source exponent. A finite number of lower monomials cannot catch a higher monomial. This is the precise reason why the extra minor comparisons do not lower the maximal order type in the range covered by the theorem.

Remark 3.5 (The boundary of the theorem). The code construction is valid for any ordinal α for which a connected linearization is available. Its lower bound is always ω^α . The upper bound supplied by multiset embedding agrees with it below ε_0 , but not uniformly at larger ordinals. Thus the restriction on α is part of the argument, not a dispensable technicality. Computing $\mathbf{o}(C)$ itself can also be the hard part.

4 Finite connected templates and linear forests

4.1 Width of a finite-dimensional grid

Lemma 4.1. *For every integer $d \geq 1$,*

$$\mathbf{o}(\mathbb{N}^d) = \omega^d, \quad \mathbf{h}(\mathbb{N}^d) = \omega, \quad \mathbf{w}(\mathbb{N}^d) = \omega^{d-1}.$$

Proof. The maximal order type follows from the natural-product theorem. The sum of the coordinates is a finite strict rank, and there is an infinite chain, so the height is ω .

For the width upper bound, use induction on d . The case $d = 1$ is a chain. For $d \geq 2$ and $x \in \mathbb{N}^d$, every point incomparable with x has some coordinate $y_i < x_i$. The incomparable residual is therefore covered by the finitely many slices

$$S_{i,t} = \{y : y_i = t\}, \quad 1 \leq i \leq d, \quad 0 \leq t < x_i.$$

Each slice is isomorphic to $\mathbb{N}^{d-1}$. By induction and equation (4), its residual has width at most $\omega^{d-2} \cdot (x_1 + \dots + x_d)$. Taking the supremum in equation (2) yields $w(\mathbb{N}^d) \leq \omega^{d-1}$.

For $d \geq 2$, suppose the inequality were strict. An ordinal $\beta < \omega^{d-1}$ has only finite Cantor-normal-form exponents at most $d - 2$. The ordinal $\omega \otimes \beta$ then has leading exponent at most $d - 1$, and is strictly below ω^d . This contradicts theorem 2.4. The case $d = 1$ already gives equality. $\square$

4.2 A formula independent of template geometry

Theorem 4.2 (Finite template theorem). *Let C contain exactly $N \geq 1$ isomorphism types of nonempty connected finite simple graphs. Then*

$$o(\mathcal{A}(C)) = \omega^N, \quad w(\mathcal{A}(C)) = \omega^{N-1}, \quad h(\mathcal{A}(C)) = \omega.$$

Proof. The finite connected order has maximal order type N , so theorem 3.3 gives ω^N . There is also a direct upper bound: record the N component multiplicities. Coordinatewise increase of this vector is sufficient for minor containment, so the graph order is an augmentation of $\mathbb{N}^N$.

Every strict minor decreases $|V| + |E|$, which supplies a finite rank to each graph. Repeating any fixed template gives an infinite chain. Thus the height is ω . The same multiplicity representation and lemma 4.1 give $w \leq \omega^{N-1}$. If $N \geq 2$, a smaller width would imply $\omega \otimes w < \omega^N$, contradicting theorem 2.4. If $N = 1$, nonemptiness gives $w \geq 1$, and the upper bound is already 1. $\square$

The surprising feature is not that a finite list has a finite connected order type, but that adding all finite disjoint unions produces precisely one factor of ω per connected isomorphism type. Allowing contractions, edge deletions, and component splitting changes the comparabilities but not this answer.

4.3 Paths alone

Let $\mathcal{L}_b$ be the minor order on finite disjoint unions of $P_1, \dots, P_b$, and let $\mathcal{L} = \mathcal{L}_\infty$ allow all finite paths.

Corollary 4.3. *For finite $b \geq 1$,*

$$o(\mathcal{L}_b) = \omega^b, \quad w(\mathcal{L}_b) = \omega^{b-1}, \quad h(\mathcal{L}_b) = \omega.$$

For unrestricted linear forests,

$$o(\mathcal{L}) = w(\mathcal{L}) = \omega^\omega, \quad h(\mathcal{L}) = \omega.$$

Proof. The finite statements are theorem 4.2. The connected path order $P_1 < P_2 < \dots$ has type ω , so theorem 3.3 gives $o(\mathcal{L}) = \omega^\omega$. Each $\mathcal{L}_b$ is an induced suborder of $\mathcal{L}$, whence

$$w(\mathcal{L}) \geq \sup_{b \geq 1} \omega^{b-1} = \omega^\omega.$$

The reverse bound is $w \leq o$. The vertex–edge norm bounds height by ω , and the isolated-vertex graphs give an infinite chain. $\square$

For a linear forest with p_n copies of P_n , a maximal linearization is explicitly

$$\Phi_P(G) = \sum_{n \text{ decreasing}} \omega^{n-1} p_n. \quad (6)$$

The sum is finite and written in decreasing exponent order. It enumerates all ordinals below ω^ω .

5 The exact minor geometry of paths and cycles

Lemma 5.1 (Connected comparisons). *For valid positive path sizes and cycle sizes at least three,*

$$\begin{aligned} P_a \preccurlyeq_m P_b &\iff a \leq b, \\ C_a \preccurlyeq_m C_b &\iff a \leq b, \\ P_a \preccurlyeq_m C_b &\iff a \leq b, \\ C_a \preccurlyeq_m P_b &\text{ never holds.} \end{aligned}$$

A source cycle that supplies a cyclic target component cannot supply any other target component.

Proof. A minor cannot have more vertices than its source. The positive cases are realized by shortening a path, contracting a cycle, or opening a cycle into a path and shortening it. A path cannot yield a cycle: deletions and contractions preserve acyclicity.

For the final assertion, retaining a cycle in a minor of C_b allows neither a vertex deletion nor an edge deletion from that component before the cycle is formed; either deletion destroys its only cycle. Equivalently, a cycle model in C_b must cover its entire cyclic traversal by its branch sets. All vertices of the source component belong to the cyclic model, so there are none available for a separate target component. $\square$

Lemma 5.2 (One path bin). *A linear forest with component sizes $a_1, \dots, a_r$ is a minor of P_n if and only if $\sum_i a_i \leq n$. The same condition describes the linear forests that are minors of C_n .*

Proof. The vertex count proves necessity. For a path, choose consecutive segments of the desired sizes, delete the edges between segments, and delete unused vertices. Adjacent segments require an edge deletion, not a spare separating vertex. For a cycle, first delete one edge to obtain P_n , then use the path construction. The empty target is allowed in both cases. $\square$

Represent a graph by the sorted lists

$$G = (p_1, \dots, p_r; c_1, \dots, c_s), \quad p_i \geq 1, c_j \geq 3.$$

The p_i are path sizes and the c_j are cycle sizes.

Theorem 5.3 (Packing criterion). *A target graph H is a minor of a source graph G if and only if both of the following can be achieved together:*

- (i) *Match each target cycle to a distinct source cycle of at least its length. A matched source cycle is reserved in its entirety.*
- (ii) *Assign each target path to a source path or to an unreserved source cycle, with the sum of assigned path sizes in each bin no greater than that source component's vertex count.*

Proof. A minor model allocates every target component to a source component. By lemma 5.1, cycles require distinct cyclic sources and consume them. The remaining target components are paths, and the vertex counts in each source component give the bin constraints. This proves necessity. Conversely, contract each reserved cycle to its matched target cycle, and use lemma 5.2 independently in every remaining source component. Their disjoint union realizes the entire target graph. $\square$

Total vertex capacity alone does not suffice. For example, $P_3 \dot{\cup} P_3$ is not a minor of $P_4 \dot{\cup} P_2$, even though both have six vertices. The two length-three path components cannot be placed into the bins of sizes four and two.

Proposition 5.4 (Greedy cycle reservation is exact). *Process the target cycles in decreasing length order, always reserving the smallest remaining source cycle that is long enough. Then apply exact bin packing to the target paths using the remaining bins. The result accepts exactly the minor pairs. If no eligible cycle exists at some step, the minor test fails.*

Proof. Consider a feasible allocation and the largest as-yet unmatched target cycle, of length t . Let f be the smallest available source cycle of length at least t , and let c be its source in the feasible allocation. Then $f \leq c$ in vertex count.

If f was unused by the cyclic targets, move the target cycle from c to f and move every path assigned to f into c . This is valid because c has at least as much capacity. If f was assigned to a different target cycle of length u , swap the two assignments. Since t is largest, $u \leq t \leq c$, so the swap is valid. Thus a feasible allocation can always be changed to use the greedy reservation. Induction proves the claim. Nothing here says that greedy *path* packing is correct; the path stage remains an exact search. $\square$

6 The unrestricted degree-two order

Let C_2 be the connected graphs of maximum degree at most two. It consists of the path chain and the cycle chain, with the cross comparisons in lemma 5.1.

Lemma 6.1. *The connected minor order satisfies $o(C_2) = \omega \cdot 2$.*

Proof. Removing every cross-comparison gives the disjoint sum of two chains of type ω . Its maximal order type is $\omega \oplus \omega = \omega \cdot 2$, which is an upper bound. The order that puts all paths first in increasing size, followed by all cycles in increasing size, is a linear extension. Its order type is $\omega + \omega$, giving equality. $\square$

Theorem 6.2. *For all finite simple degree-two graphs,*

$$o(\mathcal{D}) = \omega^{\omega \cdot 2}.$$

A maximal linearization is supplied by the explicit code

$$\Phi(G) = \sum_{n \text{ decreasing}, n \geq 3} \omega^{\omega + (n-3)} c_n + \sum_{n \text{ decreasing}, n \geq 1} \omega^{n-1} p_n, \quad (7)$$

where c_n and p_n denote component multiplicities.

Proof. Apply theorem 3.3 to lemma 6.1. The connected linearization used there labels P_n by $n - 1$ and C_n by $\omega + (n - 3)$. Equation (7) is precisely equation (5). Its possible exponents are all ordinals below $\omega \cdot 2$, and all finite coefficients are allowed. It therefore ranges bijectively over the ordinals below $\omega^{\omega \cdot 2}$. $\square$

For example,

$$G = C_5 \dot{\cup} C_3 \dot{\cup} C_3 \dot{\cup} P_4 \dot{\cup} P_1 \implies \Phi(G) = \omega^{\omega+2} + \omega^\omega \cdot 2 + \omega^3 + 1.$$

Deleting the cycle edge that turns C_5 into P_5 replaces $\omega^{\omega+2}$ by ω^4 , possibly adding it to existing path terms. That is a strict ordinal decrease, regardless of the finite multiplicities of the lower terms.

Proposition 6.3 (Height and width). *The full degree-two order has*

$$h(\mathcal{D}) = \omega, \quad w(\mathcal{D}) = \omega^{\omega \cdot 2}.$$

In fact, the ordinary well-founded rank of an individual $G \in \mathcal{D}$ is exactly $|V(G)| + |E(G)|$.

Proof. Every elementary proper minor operation strictly lowers $|V| + |E|$. Conversely, delete the edges one at a time and then delete the isolated vertices one at a time. All intermediate graphs still have degree at most two. This realizes a descending chain with exactly $|V| + |E|$ steps. Thus the point rank is the asserted integer, and the height of the entire order is ω .

Put $\lambda = \omega \cdot 2$. If $\beta < \omega^\lambda$, each exponent ξ in the Cantor normal form of β is either finite or has the form $\omega + n$ for finite n . In either case $1 \oplus \xi < \lambda$. Consequently

$$\omega \otimes \beta < \omega^\lambda.$$

If $w(\mathcal{D}) < \omega^\lambda$, the height–width inequality would therefore contradict theorem 6.2. Hence the width equals the maximal order type. $\square$

Notice that the very large code $\Phi(G)$ is *not* the point rank of G . It describes the position of G in a particular maximal linear extension. The intrinsic point rank is the finite integer $|V| + |E|$. Confusing these two rankings would erase the distinction between height and maximal order type.

7 A width-amplification lemma

We need a little more than the height–width inequality to obtain sharp widths when only finitely many cycles are permitted. A transferability argument supplies the missing factor of ω . The argument below is a direct tree-rank proof of the standard transferable-order lower bound; compare [6, Section 4.4.2].

For a finite subset F of P , write

$$P_F = P \setminus \bigcup_{x \in F} \downarrow x.$$

Definition 7.1. A well-partial-order P is *transferable for width* if $w(P_F) = w(P)$ for every finite $F \subseteq P$.

Lemma 7.2 (Finite deletion). *Suppose every principal lower set of P is finite and $w(P) = \gamma$ is infinite and additively indecomposable. Then P is transferable for width.*

Proof. Removing the lower sets of finitely many points removes only a finite set E . By equation (4),

$$\gamma = \mathbf{w}(P) \leq \mathbf{w}(P \setminus E) \oplus |E|.$$

If $\mathbf{w}(P \setminus E) < \gamma$, the right side is still strictly below γ , a contradiction. The reverse inequality follows from induced-suborder monotonicity. $\square$

Lemma 7.3 (Transferable product bound). *If P is transferable for width and $\mathbf{w}(P) = \gamma$, then for every ordinal β ,*

$$\mathbf{w}(P \times \beta) \geq \gamma \cdot \beta.$$

Consequently, for every well-partial-order Q ,

$$\mathbf{w}(P \times Q) \geq \gamma \cdot \mathbf{o}(Q).$$

Proof. Prove the first assertion simultaneously for P_F , for every finite F , by transfinite induction on β . Further finite lower-set removals from P_F have the form $P_{F \cup F'}$, so they retain width γ . The assertion is immediate for $\beta = 0$. At a limit, restrict to the suborders $P_F \times \delta$ for $\delta < \beta$ and use continuity of multiplication in its right argument: $\sup_{\delta < \beta} \gamma \cdot \delta = \gamma \cdot \beta$.

For the successor step, put $R = P_F$ and consider the top slice $R \times \{\delta\}$ of $R \times (\delta + 1)$. After any finite antichain s in that slice, let A be its set of first coordinates. Every point of

$$(R \setminus \downarrow A) \times \delta$$

is incomparable with all entries of s . Indeed, a lower-slice point cannot lie above a top-slice point, and removing $\downarrow A$ prevents it from lying below one. By the induction hypothesis, the tree of these lower-slice continuations has rank at least $\gamma \cdot \delta$.

Let $r_R(s)$ be the rank of s in the antichain tree of the top slice alone. Well-founded induction over that tree shows that the rank of s in the full product tree is at least

$$\gamma \cdot \delta + r_R(s).$$

For a leaf, use the lower-slice continuation bound. At a nonleaf, take the supremum of the same bound at each top-slice child plus one; continuity of ordinal addition in its right argument gives the claim. At the empty node this is $\gamma \cdot \delta + \gamma = \gamma \cdot (\delta + 1)$.

Finally, give Q a maximal linear extension of type $\mathbf{o}(Q)$. The product with this linear extension is an augmentation of $P \times Q$, so its width is no larger. The first assertion supplies the desired lower bound. $\square$

Corollary 7.4. *For any well-partial-order Q with $\mathbf{o}(Q) = \omega^k$, where $k < \omega$,*

$$\mathbf{w}(\mathcal{L} \times Q) = \mathbf{o}(\mathcal{L} \times Q) = \omega^{\omega+k}.$$

Proof. A finite graph has only finitely many minors up to isomorphism, so principal lower sets in $\mathcal{L}$ are finite. By corollary 4.3 and lemma 7.2, $\mathcal{L}$ is transferable and has width ω^ω . Therefore

$$\mathbf{w}(\mathcal{L} \times Q) \geq \omega^\omega \cdot \omega^k = \omega^{\omega+k}.$$

The maximal order type of the product is $\omega^\omega \otimes \omega^k = \omega^{\omega \oplus k} = \omega^{\omega+k}$, which is the matching upper bound. $\square$

8 Cycle budgets and the complete parameter spectrum

8.1 A bounded number of cycles

Let $Q_{\leq k}$ be the minor order on disjoint unions of at most k cycles, and $Q_{=k}$ the suborder with exactly k cycles. A cycle cannot be split into two cycles, so these are finite-multiset embedding orders over the chain of cycle lengths.

Lemma 8.1. *For $k < \omega$,*

$$o(Q_{\leq k}) = o(Q_{=k}) = \omega^k.$$

For $k = 0$, both orders are singletons, so this means 1.

Proof. Replace a cycle length n by $n - 2 \geq 1$. Pad with zeros and sort to obtain a nondecreasing k -tuple. The multiset embeds into another exactly when its padded coordinates are componentwise no larger. One way to see necessity is to match largest entries first: the j largest target entries need j source entries at least that large. Sufficiency follows by matching the correspondingly sorted coordinates.

Thus $Q_{\leq k}$ is an induced suborder of $\mathbb{N}^k$, and has maximal order type at most ω^k . For exactly k entries, lexicographic order by the smallest coordinate first, then the next, is a linear extension of type ω^k . This is proved inductively: fixing the first positive coordinate leaves a translated nondecreasing $(k - 1)$ -tuple, of lexicographic type ω^{k-1} , and there are ω possible first coordinates. Hence the type is $\omega^{k-1} \cdot \omega = \omega^k$. The lower bound applies to both orders. $\square$

Lemma 8.2 (The exact-cycle layer is a true product). *Let $\mathcal{D}_{b,k}$ denote the degree-two graphs with path-size bound b and exactly k cycles. For finite k ,*

$$\mathcal{D}_{b,k} \cong \mathcal{L}_b \times Q_{=k}.$$

On the entire class $\mathcal{D}_{b,k}$, the minor order is an augmentation of the product order $\mathcal{L}_b \times Q_{\leq k}$.

Proof. For source and target with the same number of cycles, the injection matching target cycles to source cycles is a bijection. Every source cycle is consumed by its cyclic target and is unavailable for path components. Thus the path part must be a minor of the source path part, independently of the cycle comparison. This proves both necessity and sufficiency for the exact-cycle layer.

For at-most- k graphs, independent comparison of the two parts is still sufficient. There can be additional minor comparisons because unmatched source cycles may be opened to supply paths. This is exactly an augmentation, rather than an isomorphism in general. $\square$

Theorem 8.3 (Finite cycle budgets). *For $b \geq 1$ finite and $k < \omega$,*

$$o(\mathcal{D}_{b,k}) = \omega^{b+k}, \quad w(\mathcal{D}_{b,k}) = \omega^{b+k-1}, \quad h(\mathcal{D}_{b,k}) = \omega.$$

For unrestricted path sizes and finite k ,

$$o(\mathcal{D}_{\infty,k}) = w(\mathcal{D}_{\infty,k}) = \omega^{\omega+k}, \quad h(\mathcal{D}_{\infty,k}) = \omega.$$

Proof. Let $\alpha_b = b$ for finite b and $\alpha_\infty = \omega$. By corollary 4.3, $o(\mathcal{L}_b) = \omega^{\alpha_b}$. The product upper bound from lemma 8.2 and the induced exact-cycle product lower bound give

$$o(\mathcal{D}_{b,k}) = \omega^{\alpha_b} \otimes \omega^k = \omega^{\alpha_b \oplus k}.$$

These are precisely the two asserted maximal order types.

For finite b , encode the path part by its b multiplicities and the cycle part by its padded k -tuple. This realizes a weaker order as an induced suborder of $\mathbb{N}^{b+k}$. Consequently $w(\mathcal{D}_{b,k}) \leq \omega^{b+k-1}$. Height is ω : the vertex–edge norm gives finite point ranks, and arbitrarily many isolated vertices give an infinite chain. The height–width inequality forces the matching lower bound, just as in theorem 4.2; if $b + k = 1$, nonemptiness supplies the lower bound 1 directly.

For unrestricted path sizes, apply corollary 7.4 to the induced exact-cycle layer. Its width is $\omega^{\omega+k}$, already the maximal order type of the whole at-most- k class. The height argument is unchanged. $\square$

The finite- b classes need not be minor-closed: opening a long cycle can produce a path exceeding the chosen path-size bound. The theorem concerns their inherited minor order, not a different relation that restricts intermediate operations. This distinction is why the proof used induced suborders and augmentations rather than assuming closure.

8.2 Removing the cycle-count bound

Theorem 8.4. *For finite $b \geq 1$,*

$$o(\mathcal{D}_{b,\infty}) = w(\mathcal{D}_{b,\infty}) = \omega^\omega, \quad h(\mathcal{D}_{b,\infty}) = \omega.$$

For unrestricted b , the answer is $o(\mathcal{D}) = w(\mathcal{D}) = \omega^{\omega \cdot 2}$ and $h(\mathcal{D}) = \omega$.

Proof. When b is finite, the connected alphabet consists of finitely many paths and the chain of all cycles. Every path in the alphabet is a minor of every sufficiently long cycle. In any linear extension, all the finitely many paths therefore occur within a finite initial portion of the cycle chain. The total order type is ω . The component theorem gives $o(\mathcal{D}_{b,\infty}) = \omega^\omega$. On the other hand, the finite-budget suborders give

$$w(\mathcal{D}_{b,\infty}) \geq \sup_{k < \omega} \omega^{b+k-1} = \omega^\omega.$$

The height remains ω .

For unrestricted paths, theorem 6.2 and proposition 6.3 have already proved the result. There is also an independent width lower bound from theorem 8.3:

$$w(\mathcal{D}) \geq \sup_{k < \omega} \omega^{\omega+k} = \omega^{\omega \cdot 2}.$$

Only monotonicity is used in these lower bounds; equality with an increasing-union supremum is not presumed. $\square$

This completes the proof of theorem 1.2.

9 Two failures of continuity

The formulas reveal why a finite enumeration, however extensive, is a poor guide to a transfinite order type unless supported by a structural argument.

9.1 An increasing union of minor-closed classes

Let $\mathcal{K}_m$ contain the degree-two graphs all of whose connected components have at most m vertices. Unlike $\mathcal{D}_{b,k}$ for finite b and unbounded cycle lengths, each $\mathcal{K}_m$ really is minor-closed. Contractions and deletions cannot enlarge a connected component.

For $m = 1$, its connected alphabet contains just P_1 . For $m \geq 2$, it contains the m paths and the $m - 2$ cycles of sizes at most m , for a total of $2m - 2$ templates. Thus

$$\mathfrak{o}(\mathcal{K}_m) = \omega^{2m-2}, \quad \mathfrak{w}(\mathcal{K}_m) = \omega^{2m-3} \quad (m \geq 2). \quad (8)$$

The finite template theorem handles $m = 1$ separately.

Corollary 9.1 (Noncontinuity even along minor-closed subclasses). *The classes $\mathcal{K}_m$ are increasing and $\mathcal{D} = \bigcup_{m \geq 1} \mathcal{K}_m$, but*

$$\sup_m \mathfrak{o}(\mathcal{K}_m) = \omega^\omega < \omega^{\omega \cdot 2} = \mathfrak{o}\left(\bigcup_m \mathcal{K}_m\right).$$

The same strict inequality holds with $\mathfrak{w}$ in place of $\mathfrak{o}$.

Proof. Every finite graph has a finite maximum component size, so the union is $\mathcal{D}$. Take the suprema in equation (8) and compare with theorem 6.2 and proposition 6.3. $\square$

The familiar continuity of ordinal exponentiation, $\sup_n \omega^n = \omega^\omega$, does not imply continuity of the operation $P \mapsto \mathfrak{o}(P)$ on increasing unions of orders. These are different operations. Although every finite bad sequence occurs in some $\mathcal{K}_m$, its continuation possibilities in the full order need not be bounded by the rank it has in that one subclass.

9.2 A discontinuity already with one cycle

Fix a finite $k \geq 1$. Then

$$\mathcal{D}_{\infty,k} = \bigcup_{b \geq 1} \mathcal{D}_{b,k}, \quad \sup_b \mathfrak{o}(\mathcal{D}_{b,k}) = \omega^\omega,$$

whereas

$$\mathfrak{o}(\mathcal{D}_{\infty,k}) = \omega^{\omega+k} > \omega^\omega.$$

The same conclusion holds for ordinal width. One allowed cycle already produces the first strict jump, to $\omega^{\omega+1}$. The exact-cycle layer explains its source: an unrestricted linear forest, of type ω^ω , varies independently alongside an unbounded cycle length, contributing another factor of ω .

9.3 A maximal code need not remain maximal on a suborder

The explicit code equation (7) is maximal on $\mathcal{D}$, but restricting it to an induced subclass can lose maximality.

Proposition 9.2. *For finite $k \geq 1$, the order induced by the code Φ on $\mathcal{D}_{\infty,k}$ has type $\omega^{\omega+1}$. For $k \geq 2$, this is strictly below $\mathfrak{o}(\mathcal{D}_{\infty,k}) = \omega^{\omega+k}$.*

Proof. There are only finitely many multisets of at most k cycles whose largest cycle length is bounded by any given integer. The cycle polynomials, ordered by their Cantor normal forms, consequently have order type ω : the largest length is primary, and there are only finitely many preceding patterns for each length bound. The empty pattern is included and does not change that type. For each fixed cycle pattern, the unrestricted path part has type ω^ω , and every graph in one pattern precedes every graph in the next. The total type is therefore $\omega^\omega \cdot \omega = \omega^{\omega+1}$. $\square$

For two cycles, sorting by the largest cycle first creates only finite cycle-pattern blocks, whereas a maximal linearization of the exact-two-cycle order sorts by the *smallest* cycle first and has type ω^2 . This concrete contrast is a useful diagnostic: a correct strictly monotone ordinal code proves a lower bound on maximal order type, but does not automatically prove that the bound is optimal.

10 Exact computation and reproducible checks

10.1 What was checked

The archive contains a dependency-free Python program, `code/verify_degree_two.py`. Its principal test enumerates *every* degree-two graph up to isomorphism with at most fourteen vertices. Isomorphism is exact here: a sorted multiset of path sizes and a sorted multiset of cycle sizes uniquely determine the graph. No graph-isomorphism heuristic is used.

Two substantially different constructions of the minor relation are compared. The first takes transitive closure under elementary minor operations on component signatures. The second implements the cycle-reservation and bin-packing criterion. A third version replaces exhaustive cycle reservation by the proved greedy reservation rule; it shares the exact path-packing solver with the second version.

The completed run produced the following results.

Quantity	Verified count
Graphs up to isomorphism, at most 14 vertices	1,500
Ordered source–target pairs compared	2,250,000
Minor pairs, including equality	378,324
Proper minor pairs with strict code decrease	376,824
Equal-cycle-count product comparisons	732,568
Individual finite height-rank checks	1,500
Targeted regression examples	9

All comparisons passed. The saved report records Python 3.13.5 and a runtime of approximately 10.1 seconds in the execution environment. Runtime is not intended as a portable benchmark. The raw counts and machine-readable report are in `results/counts.csv` and `results/verification.json`.

10.2 The elementary-operation implementation

For a path P_n , deleting an edge creates $P_i \dot{\cup} P_{n-i}$ for $1 \leq i < n$; deleting a vertex creates $P_i \dot{\cup} P_{n-1-i}$, omitting zero-size pieces; contracting an edge creates P_{n-1} . Every possible position is included.

For C_n , deleting an edge creates P_n , deleting a vertex creates P_{n-1} , and contracting an edge creates C_{n-1} when $n \geq 4$. Contracting an edge of C_3 creates P_2 , because parallel edges are suppressed in the simple-graph convention. This last case is explicitly tested; treating it as a two-cycle would silently change the mathematical category.

Every elementary successor has a smaller value of $|V| + |E|$. Memoized closure therefore terminates. Since elementary operations generate the minor relation, this construction does not assume the packing criterion it is checking.

10.3 The packing implementation

The exact path-packing routine processes target paths in decreasing size. At each step it tries every distinct residual bin capacity large enough for the next path, subtracts the path size, sorts the remaining capacities, and memoizes the resulting state. Equal capacities are symmetric and need only one representative branch. It rejects immediately when total capacity or the largest capacity is insufficient. These are necessary tests only; remaining cases are solved by exhaustive branching, not by a greedy heuristic.

The following pseudocode describes the greedy-cycle variant.

```
minor(target, source):
    available_cycles = source.cycles sorted increasingly
    for need in target.cycles sorted decreasingly:
        reserve the smallest available cycle with size >= need
        if none exists: return false
        remove that cycle from available_cycles
    bins = source.paths + available_cycles
    return exact_bin_pack(target.paths, bins)
```

The program also verifies the exact-cycle product identity directly on every tested pair with the same number of cycles. This is a separate structural assertion, not just another expression of the ordinal-code monotonicity test.

10.4 Exact ordinal comparisons, not numeric approximations

For the code Φ , a path exponent $n - 1$ is stored as $(0, n - 1)$, and a cycle exponent $\omega + (n - 3)$ as $(1, n - 3)$. A graph code is a descending list of pairs consisting of an exponent and a positive coefficient. Lexicographic comparison of these finite lists is exactly comparison of the represented Cantor normal forms.

The program checks that all graph codes are distinct and that every proper minor pair has a strict decrease. It does not assign a large floating-point number to ω , and it does not attempt to approximate transfinite arithmetic numerically.

10.5 An independent enumeration check

Let a_n be the number of isomorphism classes on exactly n vertices. Component multiplicities give the formal power series

$$\sum_{n \geq 0} a_n x^n = \prod_{j \geq 1} \frac{1}{1 - x^j} \prod_{j \geq 3} \frac{1}{1 - x^j}. \quad (9)$$

The first product chooses paths and the second cycles. A coefficient calculation by repeated unbounded-knapsack updates is checked against the partition-based graph enumeration. The resulting sequence through $n = 14$ is

$$1, 1, 2, 4, 7, 11, 19, 29, 46, 70, 106, 156, 232, 334, 482.$$

These coefficients sum to 1,500. Equation (9) is an enumeration identity derived here from the component description; it is not used to infer any of the transfinite ranks.

10.6 What the finite checks do not prove

For a fixed vertex cutoff, the graph order is finite, and its maximal order type is simply the number of graph isomorphism classes below that cutoff. A test on fourteen vertices cannot observe $\omega^{\omega \cdot 2}$. The program checks the graph-theoretic criterion, its implementation, the strict ordinal-code comparison, and the exact-cycle decomposition on a finite domain. The infinite ordinal statements depend on the written arguments and the named standard theorems, not on computational extrapolation.

No theorem prover was used for the infinite proofs. The manuscript and checker are mutually supporting artifacts, not a formal verification certificate.

11 What the attack reveals about the general problem

11.1 Disjoint unions are not always the difficult step

Suppose a connected graph class has a known maximal order type below ε_0 . The component theorem determines the maximal order type of all its finite disjoint unions without solving the exact minor-allocation problem between arbitrary disconnected graphs. The upper bound ignores splitting; the lower bound tolerates every finite split into proper connected minors. Their agreement makes the detailed splitting geometry irrelevant to this one invariant.

This observation also explains why a difficult finite decision procedure can coexist with a simple transfinite formula. Determining whether a particular collection of paths fits into particular source components is a packing problem. Determining the maximal order type of the entire family can instead be done by a multiset upper bound and a Cantor-normal-form lower bound.

11.2 A concrete next step: a bounded branching template

A natural continuation is to replace cycles by a class of connected trees with a small, fixed branching pattern but unbounded arm lengths. The immediate task would be to identify the connected minor order, including all minor operations that merge or remove the distinguished branching structure, and compute its maximal order type. If that connected type were some $\alpha < \varepsilon_0$, the disjoint-union answer would then be ω^α by the proved transfer theorem.

This is a conditional route, not a claimed calculation for such tree classes. The degree-two work identifies two features that would need to be replaced. First, a cycle retained as a cycle consumes the entire source component, creating exact product layers. A branching tree may retain its core while releasing other vertices for extra components. Second, paths and cycles have an elementary capacity description. For trees with branching, total vertex capacity alone cannot encode branch-set geometry.

11.3 Why the full graph-minor question remains outside this result

The component theorem does not calculate the connected order type for arbitrary graphs. At large ordinal parameters, its lower and upper bounds also need not agree, as noted in remark 3.5. Labeling changes the connected alphabet and may change which splitting operations are legal. Finally, the exact product argument for a fixed cycle budget relies on a property that general connected graphs do not possess.

The outcome of the present attempt is therefore an exact calculation on a nontrivial part of the graph-minor order, a general transfer mechanism in a stated ordinal range, and explicit counterexamples to a tempting continuity principle. These statements are the proved content. The general ordinal measurement program remains a separate problem.

A Proof audit and dependency map

The following audit separates the inherited theorems from the steps that carry the graph-specific calculation.

Claim	Dependency and critical check
Connected-to-disconnected transfer	Uses the maximal-linearization theorem and the below- ε_0 multi-set formula. The graph-specific step is lemma 3.1: an unchanged connected component cannot reproduce itself plus additional components.
Finite template width	Uses the grid width calculation and the height-width inequality. The multiplicity map supplies only a weaker coordinatewise order; no converse matching assertion is assumed.
Full degree-two type	The connected type is exactly $\omega \cdot 2$, not merely ω . Paths-first, cycles-second is a valid linear extension, and the incomparable disjoint-chain order supplies its upper bound.
Full degree-two width	The argument checks directly that $\omega \otimes \beta < \omega^{\omega \cdot 2}$ for every smaller β . It does not incorrectly assume that $\omega^{\omega \cdot 2}$ is multiplicatively indecomposable.
Finite cycle budget	The source and target have the same number of cycles on the lower-bound layer. Hence every source cycle is reserved; no path can use any of its unused-looking length.
Unbounded-path width	Finite principal lower sets and additively indecomposable width make $\mathcal{L}$ transferable. The product lower bound is proved by antichain-tree induction, with ordinary multiplication in the specified order.
Union discontinuity	Both the subclass invariants and the full-class invariant are calculated independently. No interchange of a union and a rank operation is used.
Computer results	Exhaustive within the stated vertex cutoff. Elementary-operation closure and packing are independent descriptions of minors; the greedy and exhaustive cycle-allocation variants share the same path-packing subroutine.

Priority and verification status. The written proofs are not formally checked or externally refereed. The broad source problem is documented, while historical novelty of the special-case answers is not established. The exact finite tests passed; their range and limits are reported rather than converted into a claim about infinite proof verification.

B Rebuilding the artifacts

The code uses the Python standard library only and is written for Python 3.9 or later. From the extracted archive root, run

```
python code/verify_degree_two.py --max-vertices 14 --out results
python build.py
```

The second command invokes `pdflatex` repeatedly and requires a TeX installation with the packages listed in the document preamble. The supplied PDF is already compiled. Alternatively,

```
latexmk -pdf -interaction=nonstopmode -halt-on-error graph_minor_ordinals.tex
```

The JSON report records the execution environment and all principal test counts. The CSV file records enumeration counts by vertex number. The source-review notes record the scope of the literature check and the exclusion of a lexicographic-product question that was found to have a later published proof.

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